

Prem Kumar Shaw

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PROFESSIONAL SUMMARY

B.Tech Computer Science (Data Science) graduate with strong skills in Python, FastAPI, SQL, and software engineering fundamentals. Experienced in building REST APIs, web applications, and AI-powered solutions through internships and personal projects. Passionate about scalable backend development, problem solving, and AI-assisted software engineering.

TECHNICAL SKILLS

Programming Languages	: Python, SQL, JavaScript, C++
Backend Technologies	: FastAPI, REST APIs, Pydantic, SQLAlchemy (ORM), JWT Authentication
Databases	: MongoDB, MySQL
Web Technologies	: HTML, CSS
Libraries & Frameworks	: PySpark, Streamlit, Scikit-learn, Pandas, NumPy
AI Tools	: ChatGPT, Claude, Gemini API
Cloud & Tools	: Git, GitHub, Docker, Power BI, VS Code, Jupyter Notebook, Google Colab
Core Concepts	: DSA, OOP, DBMS, SDLC, Operating Systems, Computer Networks, System Design (Basic)

INTERNSHIP EXPERIENCE

Forcas Studio Limited — Data Analytics Intern

Kolkata | Aug 2025 – Oct 2025

- Managed end-to-end data operations for Amazon, Flipkart, Ajio, and Meesho e-commerce platforms.
- Automated daily data processing of 800,000+ records using Python, improving pipeline efficiency by 50%.
- Designed and executed ETL operations and advanced SQL queries to derive actionable business insights.
- Built interactive Excel dashboards and Pivot Tables to support data-driven business decisions.

Techno Exponent — Machine Learning Engineer Intern

Remote | Aug 2025 – Nov 2025

- Built and optimized machine learning models for structured datasets using Python and Scikit-learn.
- Developed a CatBoost regression model using advanced feature engineering to predict diamond prices, achieving $R^2 \approx 0.97$.
- Developed backend logic and deployed ML model using Streamlit for real-time predictions.

PROJECTS

AI Resume Interview Coach (In Progress) | Python, FastAPI, MongoDB, Gemini API, JWT Authentication

- Developing an **AI-powered** mock interview platform that generates personalized interview questions based on users' resumes.
- Building **REST APIs** with **FastAPI** and integrating **MongoDB** for user data and interview history.
- Implementing AI-powered answer evaluation and personalized feedback using the **Gemini API**.
- Integrating **Speech-to-Text** and **Text-to-Speech** for voice-based interview interactions.

Smart Restaurant Management System | Python, FastAPI, Streamlit, API Documentation

Live Demo: [Link](#) | GitHub: [Link](#)

- Built and deployed a full-stack restaurant management application using **FastAPI** (backend) and **Streamlit** (frontend) on **Streamlit Cloud**, providing a publicly accessible **live demo**.
- Designed and implemented **RESTful APIs** (/menu, /order, /sales) for real-time order processing and data retrieval.
- Developed an interactive UI with a **cart system supporting multiple item orders**, improving user experience and usability.
- Implemented **state management and dynamic pricing logic**, enabling real-time total calculation and sales tracking.

EDUCATION

Brainware University — B.Tech in Computer Science & Engineering (Data Science)

CGPA: 8.57 / 10 | Percentage: 82%

Kolkata, India

2022 – 2026

CERTIFICATIONS

- Software Engineer Intern Certificate — [HackerRank](#)
- SQL (Basic & Intermediate) Certifications — [HackerRank](#)
- Artificial Intelligence — [IBM SkillsBuild](#)
- Python Certificate — [Udemy](#)

ACHIEVEMENTS

- Solved 150+ coding problems on LeetCode and HackerRank, covering Data Structures, Algorithms, and SQL.
- Solved 50+ SQL problems on LeetCode and earned SQL (Basic & Intermediate) badges.
- School Topper (**1st Rank**) in Class X (2019) and Class XII (2021).